TIDEHUNTER: Large-Value Storage With Minimal Data Relocation

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Abstract

Log-Structured Merge-Trees (LSM-trees) dominate persistent key-value storage but suffer from high write amplification from 10× to 30× under random workloads due to repeated compaction. This overhead becomes prohibitive for large values with uniformly distributed keys, a workload common in content-addressable storage, deduplication systems, and blockchain validators. We present TIDEHUNTER, a storage engine that eliminates value compaction by treating the Write-Ahead Log (WAL) as permanent storage rather than a temporary recovery buffer. Values are never overwritten; and small, lazily-flushed index tables map keys to WAL positions. TIDEHUNTER introduces (a) lock-free writes that saturate NVMe drives through atomic allocation and parallel copying, (b) an optimistic index structure that exploits uniform key distributions for single-roundtrip lookups, and (c) background relocation that reclaims space without blocking writes. On a 1 TB dataset with 1 KB values, TIDEHUNTER achieves 763K writes per second, that is 6.8× higher than RocksDB and 2.2× higher than BlobDB, while improving point queries by 2× and existence checks by 21.1×. We validate real-world impact by integrating TIDEHUNTER into Sui, a high-throughput blockchain, where it maintains stable throughput and latency under loads that cause RocksDB-backed validators to collapse. TIDEHUNTER is production-ready and is being deployed in production within Sui.

1 Introduction

Log-Structured Merge-Trees (LSM-trees) dominate persistent key-value store design. An LSM-tree buffers writes in memory, flushes them to disk as sorted files, and periodically merges (compacts) those files to bound their count and keep lookups fast. For small values, the rewriting overhead is acceptable; for values of several kilobytes, it is not. Write amplification (bytes written to disk divided by bytes from the application) reaches 10× to 30× [26] under random workloads and grows with dataset size. A store ingesting 100 MB/s of application data may push 1 GB/s to disk, starving reads of bandwidth.

LSM-trees were designed for hard disk drives, where sequential I/O is 100× faster than random and trading write amplification for sequential access made sense. Solid-state drives (SSDs) changed the equation: sequential-to-random gaps shrink to 10× or less for large requests, and SSDs expose internal parallelism that LSM-trees cannot exploit. SSD cells also wear out after a finite number of writes, so high write amplification accelerates device failure.

WiscKey [41] introduced key-value separation: keys stay in an LSM-tree, values go to a separate append-only log. Values stop

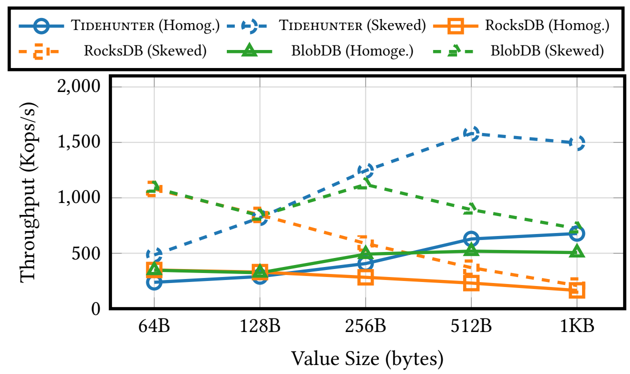


Figure 1: Mixed workload throughput (50% reads/writes) vs. value size, with skewed (Zipf $\theta = 2$) and homogeneous (Zipf $\theta = 0$) access patterns.

moving during compaction, and write amplification drops. However, the LSM-tree remains, and so do its compaction cost and multi-level lookup overhead. Under uniform key distributions, lookups traverse multiple levels with little cache benefit, and WiscKey's value log needs its own garbage collection, which stalls I/O.

Some designs pair an append-only log with a hash index to avoid LSM compaction. Bitcask [58] and FAWN-DS [2] use in-memory hash indexes for fast lookups, but their index must fit in RAM, thus bounding dataset size and slowing down startups. CedrusDB [63] uses a memory-mapped lazy-trie for on-disk access and faster recovery, but this still requires multiple disk accesses to locate keys.

We introduce TIDEHUNTER, in which the Write-Ahead Log (WAL), traditionally a temporary crash-recovery buffer, becomes the permanent store for values. We replace the key LSM-tree with sharded index tables that flush lazily. Unlike in-memory hash indexes, TIDEHUNTER caches indices on demand, sidestepping memory limits and avoiding full loads at startup. Moreover, rather than traversing an on-disk data structure, it directly estimates a key's on-disk index position.

Our design targets applications with large values and uniformly distributed keys. Content-addressable storage systems like Git and IPFS key objects by cryptographic hashes of their contents [7]. Deduplication systems for backup and archival storage partition data into chunks and index them by hash, with chunk sizes typically ranging from 4 KB to 64 KB [31, 66]. Object stores and blob storage systems often use UUIDs or content hashes as keys; Facebook's Haystack, for example, stores billions of photos keyed by unique identifiers with a size of around 8-64 KB [6]. Machine learning feature stores index embeddings and feature vectors by entity ID or feature hash, with vectors often reaching several kilobytes [21, 23, 52]. The shift toward immutable, content-addressed, and UUID-keyed data spans cloud infrastructure, data pipelines, and application backends. In all these settings, keys lack locality by design, and values are large enough that LSM-tree compaction becomes the dominant cost.

We focus on blockchain validators as a concrete evaluation target because they combine all of these pressures in a single system: hash-keyed data, kilobyte values, write-heavy ingestion, latency-sensitive reads, and aggressive pruning. We integrate TIDEHUNTER into Sui [9], a high-throughput blockchain whose validators handle thousands of transactions per second and serve concurrent queries over history, execution results, and object state. Validators write continuously as blocks arrive but must read with low latency, and they prune aggressively, keeping only recent epochs (bounded retention windows). This stresses the underlying RocksDB store, where ill-timed compactions can collapse throughput.

TIDEHUNTER is built for large values and uniform keys. Figure 1 shows its throughput advantage growing with value size, under both skewed and homogeneous (uniformly random) access. Three insights guided our design:

(1) **Separate values from indices.** Values live in a central WAL; indices are small and separate. Compaction rewrites indices, not values. Large values are written once and never copied.
(2) **Manage indices lazily.** Indices load on-demand and unload under memory pressure, enabling working sets larger than RAM without manual tuning.
(3) **Exploit uniform key distribution.** When keys are hashes or UUIDs, their distribution is known a priori. TIDEHUNTER estimates a key's on-disk index position directly from the key bits, resolving lookups in a single I/O without trie or tree traversal and without training a model.

We evaluate TIDEHUNTER against RocksDB and BlobDB (the production implementation of WiscKey) on 1 TB datasets. At 1 KB values, TIDEHUNTER achieves 763K writes/sec (up to 6.8× faster than RocksDB and 2.2× faster than BlobDB), improves point queries by 2.0×, and speeds up existence checks by 21.1× (resolved from the index, no value fetch). For small 64-byte values on homogeneous workloads, LSM-trees remain 1.4× to 2.2× faster.

To validate that synthetic benchmarks translate to real systems, we run Sui validators under sustained transaction load. With RocksDB, validators degrade at around 6,000 transactions per second as disk utilization rises and latency climbs. Under the same load, TIDEHUNTER maintains stable throughput and lower latency with reduced disk pressure, avoiding the I/O saturation that triggers RocksDB's performance collapse.

Contributions. This paper makes the following contributions:

- We present TIDEHUNTER, a storage engine architecture that eliminates value compaction by treating the WAL as permanent storage, achieving near 1× write amplification for large values.
- We introduce an optimistic index that exploits uniform key distributions to achieve single-roundtrip lookups, trading the generality of LSM-trees for lower tail latencies.
- We evaluate TIDEHUNTER on synthetic benchmarks and a production blockchain, demonstrating up to 6.8× higher write throughput compared to RocksDB for 1 KB values and stable performance where RocksDB-backed systems collapse.

2 Background & Design Principles

We review LSM-trees and their compaction costs, then derive the principles guiding TIDEHUNTER's design.

2.1 Log-Structured Merge-Trees

LSM-trees [50] emerged as the dominant architecture for write-intensive key-value stores, exploiting a fundamental disk-drive asymmetry: sequential I/O is roughly 100× faster than random [32, 57]. Rather than updating records in place (which requires random seeks), LSM-trees buffer writes in memory and flush them sequentially, converting random writes into sequential at the cost of extra bookkeeping.

An LSM-tree organizes data into levels. Incoming writes land in an in-memory *memtable*; when full, the engine flushes it to disk as an immutable Sorted String Table (SST) file. These files form Level 0 (L0) and may have overlapping key ranges since each flush produces an independent file. As L0 accumulates files, read performance degrades: a lookup checks the memtable, then each L0 file in turn. To bound this, periodic *compaction* merges L0 files into Level 1 (L1). L1 and deeper levels maintain a key invariant: within a level, files have non-overlapping key ranges, so a lookup at L1 or beyond touches a single file per level. When L1 grows too large, files merge into L2, and so on.

RocksDB [26] is a widely deployed implementation that uses leveled compaction by default. Each level is roughly 10× larger than the previous, so a dataset of size N requires $O(\log N)$ levels.

2.2 Compaction Costs

Compaction maintains read performance but introduces three forms of overhead that dominate under write-heavy workloads.

Write amplification. When compaction merges files from Li into L$i+1$, every record is rewritten even if unchanged. With 10× size ratios between levels and $O(\log N)$ levels, write amplification (bytes to disk divided by bytes from the application) reaches 10× to 30× under random workloads [26]. In Sui's private testnet, validators ingesting 40 MB/s of application data generated 500 MB/s of disk writes (an 12.5× amplification). High write amplification consumes I/O bandwidth, accelerates SSD wear [33], and limits throughput scaling.

Read amplification. Reads still traverse multiple levels: a point query may touch the memtable, one or more L0 files, and one file per deeper level. Bloom filters mitigate negative lookups [26], but positive lookups on cold data incur one disk read per level (five or six for a 1 TB dataset with 10× level ratios).

Write stalls. Compaction competes with foreground writes for I/O and CPU. To keep L0 bounded, RocksDB throttles or blocks writes when compaction falls behind, causing *write stalls* that produce latency spikes and throughput drops [5, 26]. The result is unpredictable performance: the same workload may sustain high throughput one minute and stall the next. These costs compound

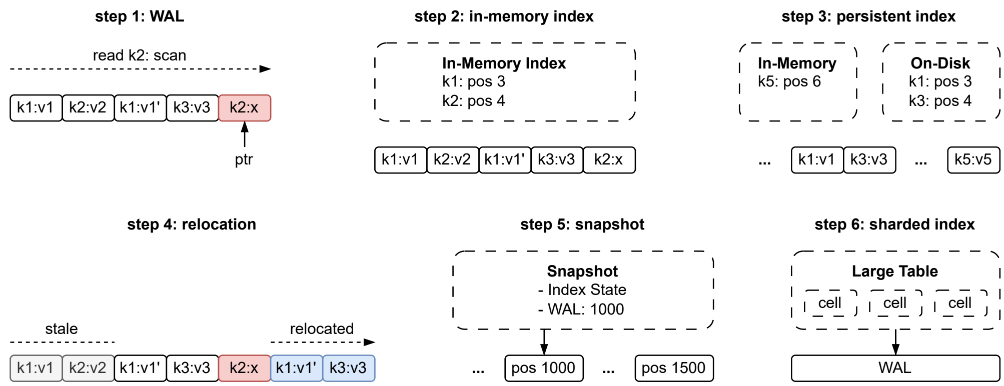


Figure 2: Design evolution of TIDEHUNTER. Each step addresses a limitation of the previous design: (1) append-only log provides fast writes but slow reads; (2) in-memory index enables fast lookups; (3) persisted index supports data larger than RAM; (4) relocation reclaims space without blocking writes; (5) snapshots enable fast recovery; (6) sharding improves concurrency.

with dataset and value size. In Sui's testnet, disk I/O utilization approached 80% with no headroom to scale throughput; the bottleneck was not the application but the storage engine rewriting unchanged data. TIDEHUNTER eliminates value compaction entirely: values land in the WAL once and never move again.

2.3 TIDEHUNTER Design Principles

These limitations, particularly acute for applications with large values and uniform keys, motivate TIDEHUNTER's design. Figure 2 presents our journey towards TIDEHUNTER by starting from a minimal storage system and incrementally addressing its limitations. Each step introduces a component that appears in the final architecture (Figure 2). Three principles guide every decision: (1) writes append directly to their long-term position on disk, (2) values are rarely moved, and (3) compaction never blocks writes.

Step 1: Append-Only Log. The simplest design appends all writes (with tombstones for deletes) to a single log. Reads scan backward from the tail, returning the first match. This achieves $O(1)$ writes but $O(n)$ reads, acceptable for write-only workloads but impractical otherwise.

Step 2: In-Memory Index. An in-memory index mapping keys to log positions reduces reads to $O(\log n)$ or $O(1)$ while writes still append in one I/O. The index grows with the number of unique keys, so for datasets larger than memory it becomes the bottleneck.

Step 3: Persisted Index. We periodically flush the in-memory index to disk as a sorted file. Reads check the in-memory index for recent writes, then fall back to the persisted index. This handles datasets larger than RAM, but the WAL grows unbounded: deleted keys and overwritten values consume space forever.

Step 4: Relocation. A *relocation* process moves live entries forward in the log and discards obsolete ones, scanning the oldest entries and re-appending those still live to the tail; once a segment is empty, it can be deleted. Crucially, relocation runs in the background and never blocks writes: unlike LSM compaction, the relocator just appends to the same log that serves foreground writes. Recovery, however, still requires replaying the entire WAL to reconstruct the index, which is impractically slow for large datasets.

Step 5: Snapshots. Periodic snapshots record the index state and a WAL position on disk; recovery loads the snapshot and replays only the WAL suffix after it. Snapshots store only metadata (index positions and WAL offsets), not the index data itself, so they are small and cheap to take frequently. A single index, however, becomes a contention point under concurrent writes.

Step 6: Sharded Index. Partitioning the key space into independent cells, each with its own in-memory index, persisted index, and lock, lets operations on different shards proceed in parallel. The WAL remains shared (writes serialize only at allocation), but index operations parallelize across shards. §3 describes each component in detail.

3 The TIDEHUNTER Architecture

Figure 3 illustrates TIDEHUNTER's architecture. This section describes how its components interact to serve reads, writes, and how it does crash recovery.

The system comprises several main components:

- **Large Table**: An in-memory index mapping keys to positions in the *Value WAL*, an append-only log that permanently stores key-value entries.
- **Value WAL Controller**: Manages WAL writes, handling position allocation, memory-mapped I/O, and background durability.
- **Index Store**: An on-disk log storing serialized indices flushed by its controller under memory pressure.
- **Snapshot Engine**: Persists a point-in-time image of the Large Table to the *Control Region* for crash recovery.
- **Relocator**: Reclaims Value WAL space in the background by migrating live entries and discarding obsolete ones.

3.1 Write Flow

Figure 4 expands the Value WAL Controller, separating the fast synchronous path (solid arrows) from asynchronous background

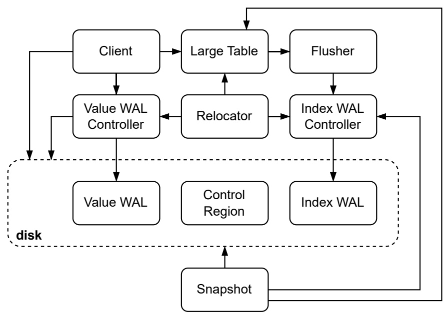


Figure 3: High-level architecture of TIDEHUNTER.

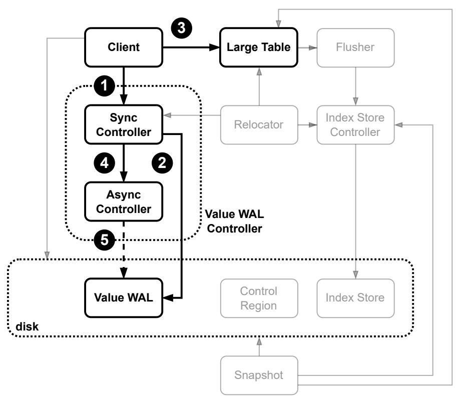


Figure 4: Write path separating fast synchronous operations (solid) from background durability (dotted).

operations (dotted). The design minimizes latency while providing clear durability guarantees.

The Client flow. The client sends a write operation (insert or delete) to the synchronous controller, which allocates a position in the Value WAL (a contiguous region for the entry) so concurrent writers do not overwrite each other (❶). Next, the client writes the entry directly to the memory-mapped Value WAL (❷). It then updates the Large Table, the in-memory index mapping keys to WAL positions, so subsequent reads can locate the value without scanning (❸). Because the allocation uses a fast atomic counter, multiple clients run the rest of the write flow in parallel.

When the update completes, the asynchronous controller is notified that the position is fully processed (entry written to the WAL and reflected in the index) (❹). It tracks the highest WAL offset whose preceding entries have all been processed, so the snapshot engine knows which WAL prefix is represented in the Large Table and crash recovery only replays entries beyond it (see §3.4).

The Value WAL is organized into memory-mapped regions called maps (also called fragments), each covering a contiguous range of positions. The asynchronous controller serves three purposes. First, it manages map lifecycle: it maintains a buffer of pre-allocated maps so writes never block on map creation, and finalizes each map once full. Second, it fsyncs finalized maps in the background (❺), keeping this blocking operation off the write path. Third, when relocation advances the minimum required WAL position, it deletes obsolete WAL files no longer needed for recovery (see §4.4).

Durability Guarantees. This design cleanly separates the synchronous write path (allocation, memory-map write, and index update) from background durability. The failure semantics are:

- **After ❷** (memory-map write): The entry is in the OS page cache, but not yet indexed. A process crash is safe because the kernel eventually flushes the page cache, allowing WAL replay to rebuild the index. However, a system crash (e.g., power loss) might lose the data because it is not fsynced.
- **After ❸** (Large Table update): The write returns to the caller. The entry is now *committed*: visible to reads and durable against process crashes, but not system crashes.
- **After ❺** (background flush): The asynchronous controller has fsynced the map. The write is now fully durable against both process and system crashes.

Providing process-crash durability at commit time and system-crash durability asynchronously matches the default behavior of RocksDB [44]. Applications needing immediate system-crash durability can request an explicit flush.

Deletes. Deleting a key writes a tombstone value. On concurrent operations (e.g., a delete and insert for the same key), the operation with the higher WAL position wins, ensuring consistency regardless of Large Table application order. Tombstones serve three purposes: they survive crash recovery (replaying the tombstone removes the key), they enable conflict resolution with concurrent writes, and they let the relocator skip deleted entries when compacting (§4.4).

Atomic batch writes. A batch is accumulated in memory and committed in two phases: a batch-start marker is written to the Value WAL followed by all entries (all under a single Value WAL allocation), and then each entry is applied to the Large Table in sequence. If a crash interrupts after the WAL write but before the index update, recovery replays the batch; the batch-start marker records the expected entry count, so incomplete or corrupted batches are safely discarded for atomic all-or-nothing visibility.

3.2 Read Flow

Figure 5 expands the Large Table from Figure 3: reads follow a tiered lookup strategy.

A read first consults the in-memory LRU cache of recently accessed values (❶); a hit returns immediately, requiring no index lookup or WAL access. On a miss, the read checks a Bloom filter (❷). If it indicates the key is absent, the lookup returns immediately, unlike RocksDB-style LSMs which traverse multiple SSTable levels regardless. Otherwise, the read consults the Large Table (❸). If the relevant index is in memory, the lookup completes immediately; otherwise it loads from the Index Store on disk (❹). The index yields a WAL position, the value is read directly from the Value WAL (❺) and cached in the LRU. In practice, frequently accessed data stays in the OS page cache, making most reads memory-speed.

When a cell's index lives on disk, loading the full index just to fetch one key is wasteful for large cells. TIDEHUNTER instead

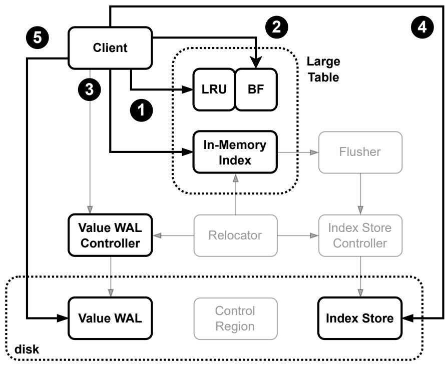


Figure 5: Read path showing the tiered lookup strategy. The snapshot component is omitted for clarity.

exploits the workload's uniform key distribution (keys are cryptographic hashes, or prefixed by such hashes) to estimate where a key would appear in the sorted serialized index: treat the key as an integer, compute its fraction of the full key range (a key starting with byte 0x80 sits roughly at the midpoint), and multiply by the index file size to get an estimated byte offset.

The system reads a small window around this offset. If the target falls inside, binary search locates it; otherwise the window shifts based on key comparison and the search repeats. For uniformly distributed keys, this typically converges in 1–3 iterations, reading only a few kilobytes rather than megabytes. We call this the *optimistic index* (it optimistically assumes the key lies within the estimated window, paying extra I/O only when the estimate is off); §4.2 details its statistical foundation. The technique is especially valuable during recovery and cold-start, when most cells are unloaded: TIDEHUNTER can serve reads immediately after opening the database, without waiting for indices to load (see §3.4).

3.3 State Snapshots

TIDEHUNTER periodically snapshots state to bound recovery time. For each cell, a snapshot stores the Index Store position of its most recently flushed index (if any) and a replay position indicating how much of the Value WAL is reflected in that flushed index. The snapshot also records a global replay-from position (the minimum replay position across all cells), which marks where Value WAL replay begins during recovery. A background thread periodically flushes cells that have accumulated significant updates and writes the snapshot metadata atomically to the Control Region, a dedicated on-disk region (Figure 3). More frequent snapshots reduce recovery time but increase write I/O; TIDEHUNTER balances this by flushing only cells whose buffered unpersisted writes exceed a configurable threshold (cell states are detailed in §4.1). Unlike LSM-trees [44] that snapshot full data structures, TIDEHUNTER only snapshots positions, which are small and cheap to rewrite frequently.

3.4 Crash Recovery

On startup after a crash or shutdown, TIDEHUNTER first reads the Control Region for the most recent snapshot (per-cell Index Store positions and the global replay-from position). It opens the Value WAL and the Index Store; both are append-only logs whose contents survived the crash intact. The Large Table is initialized from the snapshot metadata: each cell starts unloaded but knows where its serialized index resides, so indices load on demand rather than upfront. Starting from the replay-from position, TIDEHUNTER iterates through the Value WAL and replays each entry into the Large Table to reconstruct post-snapshot updates. Finally, it starts the background components (Value WAL Controller, Index Store Controller, flusher threads, relocator, snapshot thread); the database is now ready to serve clients.

4 Resource Management

This section describes how TIDEHUNTER scales beyond available memory: the Large Table organization (partial on-disk residency), index flushing (relieving memory pressure), and relocation (reclaiming disk space without blocking writes).

4.1 The Large Table

The Large Table is TIDEHUNTER's in-memory index, mapping keys to Value WAL positions. Portions of the index may reside on disk, and fine-grained locking supports concurrent access.

Partitioning and concurrency. Two challenges face an index over a dataset larger than memory: the index cannot fit entirely in RAM, and a single lock would serialize all operations. TIDEHUNTER partitions keys into cells, each spanning a contiguous key range, that can be independently loaded, evicted, and locked. Applications define key spaces at database creation, each with a distribution type. Uniformly distributed keys (cryptographic hashes) use a fixed-size cell array; keys with common prefixes use a dynamically growing B-tree, suiting composite keys like (tenant_id, record_id) and enabling efficient prefix scans without pre-allocation. Within each key space, cells are grouped into rows protected by sharded mutexes, allowing concurrent operations on different key ranges to proceed without contention.

Cell states. A naive approach would load an entire cell's index into memory on first access, but this is prohibitive for write-heavy workloads: a single write to a cold cell could trigger loading megabytes of index data. TIDEHUNTER avoids this with a *DirtyUnloaded* state that buffers new writes in memory without loading the existing on-disk index. Each cell maintains one of five states: *Empty* (initial); *Loaded* (fully resident in memory); *Unloaded* (evicted, index only on disk); *DirtyLoaded* (in-memory index with unpersisted changes); and *DirtyUnloaded* (recent writes buffered in memory while the bulk of the index remains on disk). A write to an Unloaded cell transitions it to DirtyUnloaded and buffers only the new entry. Reads check the in-memory buffer first; misses do a point lookup into the on-disk index without loading the full structure (§3.2), trading higher per-lookup cost for lower memory consumption.

4.2 Optimistic Index Lookup

When a cell's index lives on disk, lookups typically require loading the full index or using a multi-level structure with multiple I/Os. TIDEHUNTER avoids both by exploiting uniform key distributions: a key's position in a sorted array can be estimated from its value.

Consider N keys sampled uniformly from $[0, 2^{256})$ and sorted into an array. For a key k, its expected position is $k \cdot N/2^{256}$. More precisely, the i-th order statistic of N uniform samples follows a Beta distribution, $\text{Beta}(i, N - i + 1)$, which provides confidence intervals for the key's location. This statistical foundation allows TIDEHUNTER to bound the search region with high probability.

The on-disk index is a sorted array of fixed-size entries (32-byte key + 12-byte WAL position = 44 bytes), with no headers or directory structures. This enables direct position estimation: for a key k in an index of N entries, the estimated byte offset is $(k/2^{256}) \times 44N$.

To perform a lookup, the system reads a window of W entries (800 by default, about 32 KB) centered at the estimated offset. If the target falls inside, binary search locates it; otherwise the window shifts based on key comparison and the search repeats. For uniformly distributed keys, this typically converges in one to three iterations. Modern SSDs exhibit a batch-read property: reading 32 KB costs essentially the same as reading a byte, due to internal page sizes and command overhead. A window of 800 entries achieves $> 99\%$ containment probability for indices of 10K–100K entries. The window size trades hit rate against per-iteration I/O volume; we evaluate this in §6.

4.3 Index Flushing

When a cell accumulates too many dirty entries, it must flush them to disk. Blocking the cell during flush would stall write-heavy workloads, so TIDEHUNTER flushes asynchronously: the flusher snapshots the dirty index and dispatches persistence to background threads while the cell remains available for new writes.

The flusher's action depends on cell state: for DirtyLoaded cells it serializes the complete in-memory index; for DirtyUnloaded cells it first loads the existing on-disk index, merges in the dirty entries, and serializes the combined result. It may also drop obsolete entries. The prepared index is handed to the Index Store Controller for persistence. The Index Store shares the same append-only implementation as the Value WAL (§3.1), differing only in the data stored (serialized indices rather than key-value entries). The controller notifies the Large Table with the new index position, and the cell unmerges: entries included in the flush are removed from the in-memory buffer, leaving only those that arrived after flush start. If none remain, the cell becomes Unloaded; otherwise it stays DirtyUnloaded with the updated on-disk pointer.

Reads during a flush are safe because the Index Store is append-only: new indices are appended rather than overwriting. The cell keeps pointing to the previous index position until the flusher atomically swaps the pointer after the new index is written. Readers and writers thus operate on disjoint regions, requiring no coordination beyond the pointer swap.

4.4 Relocation

As writes accumulate, obsolete entries (overwritten values and deleted keys) consume Value WAL space that simple truncation cannot reclaim. *Relocation* reclaims this space by moving live entries forward in the WAL, allowing old files to be deleted: scan the WAL, identify still-live entries, re-write them at the tail, update the Large Table to point to the new positions, and delete the old files. The challenge is correctness under concurrent writes: a relocated entry must not overwrite a newer update. TIDEHUNTER uses a compare-and-set mechanism. When relocation reads an entry at position P and prepares to relocate it to P', it captures the last processed value L (the highest contiguous WAL position whose preceding writes have all been written and indexed). When applying the update, it only replaces the index entry if the current position is less than L. If a concurrent write updated the key to position $P'' > L$, the relocation update is ignored and the index continues to point to P''. This lets relocation run concurrently with normal writes without locks. Space is reclaimed at file granularity. The Value WAL is organized as a sequence of files; once all live entries in a file have been relocated (or were already beyond the file's range), the file can be deleted. The GC watermark tracks the oldest position that may still be referenced, preventing premature deletion.

TIDEHUNTER supports two relocation strategies. *WAL-based relocation* scans the Value WAL sequentially from the oldest entry; for each, it checks whether the index still references this WAL position, and if so relocates the entry. This is simple but reads every entry from disk. *Index-based relocation* iterates through Large Table cells, loads each index, identifies entries whose WAL positions fall below a configurable cutoff, reads their values, and decides which to relocate. This is more efficient when only a subset of cells contain old entries. Both strategies support an optional application-provided filter that decides for each entry: keep, remove, or stop relocation. This enables application-specific compaction logic, such as retaining only the latest version of each key. An important use is *epoch-based garbage collection*, the pattern used by Sui and other systems that organize data into time windows or versions. The application groups entries by epoch and maintains a retention boundary (the oldest epoch it must still serve); older entries can be discarded. The filter returns `Remove` if the entry's epoch is below the boundary, otherwise `StopRelocation`. Because epochs commit in order, the WAL is partitioned into contiguous epoch-regions in append order: once the scan reaches a still-retained epoch, every later entry is too. `StopRelocation` ends the pass at exactly this boundary, without reading the rest of the WAL.

A key architectural difference from LSM-tree databases is the role of compaction. In LSM-trees, compaction is essential for read performance: without it, reads search multiple SSTable levels and read amplification grows with level count. In TIDEHUNTER, read performance is independent of WAL fragmentation: the Large Table index provides constant-cost access to any value (one index lookup plus one WAL read) regardless of where in the WAL it lives. Relocation thus serves a single purpose: reclaiming disk space. Each value is written to the WAL only once during normal operation; additional writes occur only when relocation runs, and even then only on live entries. Where LSM-tree compaction trades write amplification for read performance, TIDEHUNTER's relocation trades it purely for disk space, and applications with ample storage can defer or skip it without affecting read latency.

5 Implementation

Our TIDEHUNTER implementation is *production-ready*, written in Rust [47] (about 14,000 lines, excluding tests), and being deployed to the Sui blockchain. Key libraries are: *memmap2* for memory-mapped I/O, enabling direct WAL-fragment writes without explicit system calls; *parking_lot* for high-performance mutexes with Arc-based guards used in position tracking; *arc-swap* for atomic Arc-pointer swaps enabling lock-free reads of shared structures like the WAL's memory-map registry (its set of active memory-mapped regions); *crc32fast* for WAL-entry checksums; *bloom* for per-cell Bloom filters; *lru* for the value cache; and *rayon* for parallel Large Table initialization during recovery. Unsafe code is confined to memory-mapped I/O within the WAL module. The codebase includes 160 unit tests, stress tests with shadow-state verification, and a failpoint infrastructure for deterministic concurrency testing.

TIDEHUNTER uses a fixed set of background threads. Each WAL Controller (Value and Index) runs three: a *tracker* monitoring position completion, a *mapper* managing memory-map lifecycle and GC, and a *syncer* issuing fsync on finalized fragments. A configurable pool of *flusher* threads handles index persistence concurrently across cells. A *relocator* thread performs background compaction, and a *snapshot* thread periodically checkpoints the Large Table. A typical deployment runs 8 background threads plus the configured flushers; we deliberately avoid per-request thread spawning to keep scheduling overhead and resource usage predictable.

These background threads minimize interference with client requests. Foreground threads dispatch work (index flushes, fsync, map finalization, relocation) to background threads via non-blocking message queues and return immediately. Row-level locks are held only for brief in-memory index updates and are released before any I/O or cross-thread handoff. Background threads operate on reference-counted snapshots and use atomic pointer swaps so readers see a consistent view without coordination. The relocator updates the index via compare-and-set against the last-processed position (§4.4) rather than holding cell locks, running fully concurrently with foreground writes. To bound memory when index persistence falls behind ingestion, the flusher applies back-pressure: when the number of in-flight index flushes exceeds a threshold, the foreground thread requesting the flush sleeps briefly (outside any row lock) until pending flushes drain, trading a small throughput dip for bounded memory.

6 Experimental Evaluation

We answer the following questions about TIDEHUNTER:

(1) How does the read-write ratio influence the performance of TIDEHUNTER?
(2) How does the type of read operation (get, exists, scan) influence the performance of TIDEHUNTER?
(3) How does workload skew influence the performance of TIDEHUNTER?
(4) How does value size influence the performance of TIDEHUNTER?
(5) How does relocation influence the performance of TIDEHUNTER?
(6) How does our optimistic index perform against the naive index?

6.1 Experimental Setup

We run our experiments on a Medium v4 OpenMetal instance [51] equipped with two Intel Xeon Silver 4510 processors providing 24 CPU cores (48 threads) at 2.4–4.1 GHz, 256 GB of DDR5 4400 MHz RAM, and a 6.4 TB NVMe drive.

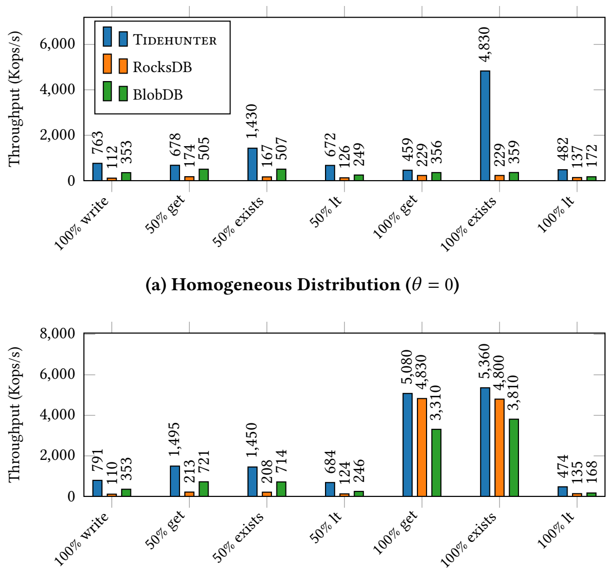
Baselines. We compare TIDEHUNTER against the following state-of-the-art key-value stores:

- **RocksDB** [44]: An LSM-tree-based embedded key-value store widely used in production systems.
- **BlobDB** [42]: RocksDB's integrated blob storage extension inspired by WiscKey [41], which separates large values from the LSM-tree to reduce write amplification.

Resource Parity and Configuration. We allocate equivalent resources to all systems to ensure a fair comparison. All three systems use the same client threads (36 write threads and 36 mixed read/write threads) and the same number of background threads: TIDEHUNTER uses 12 flusher threads, while RocksDB and BlobDB are configured with `increase_parallelism(12)`. For memory, TIDEHUNTER uses up to 128 GiB of memory-mapped WAL in total (64 fragments of 1 GiB each per WAL, applied to both the Value WAL and the Index Store), whose physical residency is managed by the OS page cache. RocksDB and BlobDB use a 128 MB block cache plus the OS page cache for SST files. On our 256 GB machine, both designs therefore rely on the OS page cache as a cache layer. We verified that increasing RocksDB's block cache to 128 GB with direct reads (to avoid double-caching) degrades throughput by 5–16% across all workloads, confirming that our smaller-cache configuration is already favorable to RocksDB. Our RocksDB compression (LZ4 with ZSTD at the bottommost level), 16 KB block size, 10-bit Bloom filters, and pinned L0 index/filter blocks follow Meta's RocksDB tuning recommendations [45], and our BlobDB settings follow the BlobDB guidance [43]. Remaining values (write buffer sizes, parallelism, L0 triggers) were selected by tuning on our 1 TB workload. Finally, all three systems use background fsync (`sync=false` for RocksDB/BlobDB, `sync_flush: false` for TIDEHUNTER), matching RocksDB's default.

Methodology. Our benchmark has two phases. In the *fill phase*, we insert random keys and values until the database reaches 1 TB (unless specified otherwise); we use a fixed 32-byte key, and the value size is a parameter. §6.2.3 additionally evaluates smaller key/-value combinations drawn from production workload studies. After fill, we allow a 10-minute cooldown for the system to stabilize (e.g., for relocation to settle). The *measurement phase* then runs operations for 10 minutes. Following YCSB's convention [17], lookups always target keys present in the database, so reads do not produce negative lookups. The following workload parameters are configurable:

- **Read-write ratio**: We evaluate three configurations: 100% writes, 50% reads with 50% writes, and 100% reads.
- **Read operation type**: We consider three read operations: *get* (returns the value for a key if it exists), *exists* (returns whether a key exists), and *reverse iterator* (returns the key-value pair with the largest key smaller than the query key, if one exists).
- **Skew**: A homogeneous workload selects keys uniformly at random, while a skewed workload favors more recently inserted keys. We implement skew using a Zipfian distribution with parameter $\theta = 0$ (homogeneous) or $\theta = 2$ (skewed). We use "homogeneous" rather than "uniform" for access patterns to avoid confusion with uniformly distributed keys.
- **Value size**: We evaluate both large values (1 KB) and small values (64 bytes and 128 bytes).



(a) Homogeneous Distribution ($\theta = 0$)

(b) Skewed Distribution ($\theta = 2$)

Figure 6: Throughput comparison across workloads. The 50% read cases include 50% writes. Top: homogeneous access distribution ($\theta = 0$). Bottom: skewed distribution ($\theta = 2$).

- **Relocation**: Relocation can be enabled or disabled to measure its impact on performance. Unless stated otherwise, all benchmark results were obtained with relocation enabled.

6.2 Benchmark Results

We present benchmark results for TIDEHUNTER, RocksDB, and BlobDB across various workload configurations.

6.2.1 Large Values. The results for large values are in Figure 6.

Write Performance. Under write-only workloads (Figure 6a, leftmost bars), TIDEHUNTER outperforms RocksDB by 6.8× and BlobDB by 2.2×. This advantage stems from TIDEHUNTER's append-only WAL design: each write requires only a single sequential disk operation, whereas LSM-tree-based systems must eventually compact and rewrite data multiple times. BlobDB partially mitigates this overhead by separating large values from the LSM-tree, explaining its intermediate performance.

Read Performance. For homogeneous reads, TIDEHUNTER maintains a consistent advantage across all operation types. On get operations (Figure 6a), TIDEHUNTER outperforms RocksDB by 2.0×; on exists, by 21.1×, because TIDEHUNTER resolves existence queries directly from the index without fetching the value, while LSM-trees must traverse multiple levels regardless of whether the value is needed. At this dataset size, TIDEHUNTER's ≈48 GB on-disk index (1.1 B records × 44 B/entry) fits entirely within the 64 GiB Index Store mmap window, so exists is a pure in-memory Bloom-filter plus index lookup; on a larger dataset that evicts the index, it would fall back to the optimistic index's 1–3-page disk-read path (§4.2). For reverse iterator operations (Figure 6a), TIDEHUNTER maintains a 3.5× advantage over RocksDB.

Mixed Workloads. At a 50/50 read-write ratio, TIDEHUNTER leads RocksDB by 3.9× on get operations. The gap narrows relative to write-only workloads but remains substantial, showing that TIDEHUNTER's design also benefits mixed read-write paths.

Effect of Skew. Under skewed workloads (Figure 6b), where accesses favor recently inserted keys, all systems benefit from caching effects. For read-only get operations with skew, TIDEHUNTER and RocksDB land within 5% of each other, with neither backend holding a meaningful lead. Hot data resides in memory across all systems: RocksDB's block cache and TIDEHUNTER's Large Table both serve frequently accessed entries without disk I/O, and the two designs converge on this workload. However, TIDEHUNTER maintains its advantage for write-heavy and mixed workloads even under skew, as its write path remains fundamentally more efficient.

Summary. For large values, TIDEHUNTER consistently outperforms LSM-tree-based alternatives across write-heavy and mixed workloads. The advantage is most pronounced for write-heavy workloads (up to 6.8×) and existence checks (up to 21.1×), and remains significant for read-heavy homogeneous workloads (2.0×). Under highly skewed 100% read workloads, TIDEHUNTER and RocksDB are within 5% of each other, with TIDEHUNTER retaining its lead in all other regimes. These results validate TIDEHUNTER's design goal of minimizing data movement.

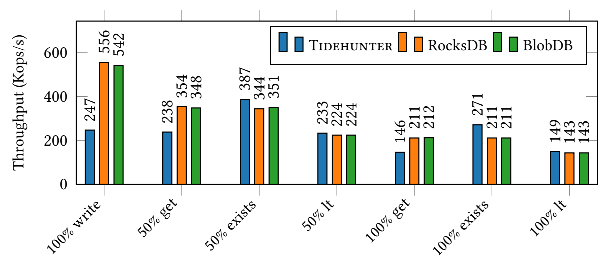
Performance Stability. To demonstrate that TIDEHUNTER's background threads do not meaningfully interfere with foreground operations, we ran 6 workload configurations (3 read ratios × 2 skew settings) for 30 min each on a single node with 1 KB values; Table 1 reports the results. Three observations establish the claim. First, foreground tail latency is tightly bounded even while the flusher and relocator run continuously: write p999 stays under 0.3 ms and read p999 under 1.3 ms across every configuration; background work does not stall foreground operations. Second, foreground lock contention on Large Table row mutexes is negligible: in five of six configurations it accounts for at most 0.001% of mean operation latency. The only outlier (100% reads, $\theta = 2$) reaches 0.07%, and even that reflects foreground readers colliding on the small set of hot cells rather than background activity; TIDEHUNTER's background threads are not the source even when contention is visible in the data. Third, throughput is steady: the coefficient of variation stays at or below 0.12 in every configuration. This variance comes from TIDEHUNTER's flusher back-pressure mechanism (§5) which briefly throttles foreground writes when pending flushes saturate.

6.2.2 Small Values. The results for small values are in Figure 7. For small values, the performance characteristics differ substantially from the large-value case, revealing the trade-offs inherent in TIDEHUNTER's design.

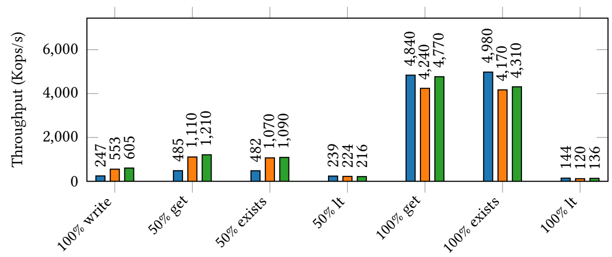
Write Performance. Under write-only workloads (Figure 7a, leftmost bars), RocksDB and BlobDB outperform TIDEHUNTER by 2.2×. This reversal occurs because the overhead of TIDEHUNTER's separate index becomes proportionally more significant when values are small. LSM-trees amortize their metadata costs across sorted runs and benefit from efficient compression of collocated entries.

Table 1: Stability across 6 workload configurations on 1 KB values. Throughput CV is the worse of write- and read-throughput CVs. "LT ovhd" is the expected per-operation wait on Large Table row mutexes, reported as a fraction of mean operation latency.

			Latency p50/p99/p999 (μs)		
R%	θ	Tput CV	Write	Read	LT ovhd %
0	0.0	0.07	28/68/234	–	0.0004
50	0.0	0.12	9/37/67	93/465/1196	0.0001
100	0.0	0.07	–	86/538/1292	<0.0001
0	2.0	0.07	28/69/270	–	0.0004
50	2.0	0.08	24/58/193	3/20/140	0.0010
100	2.0	0.006	–	4/15/19	0.068



(a) Homogeneous Distribution ($\theta = 0$)

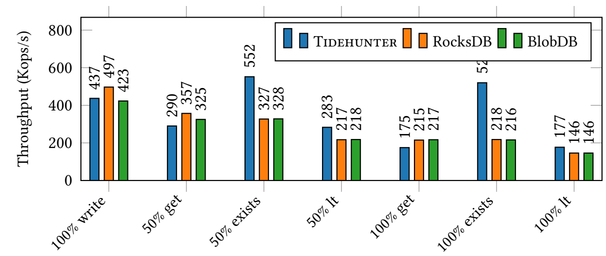


(b) Skewed Distribution ($\theta = 2$)

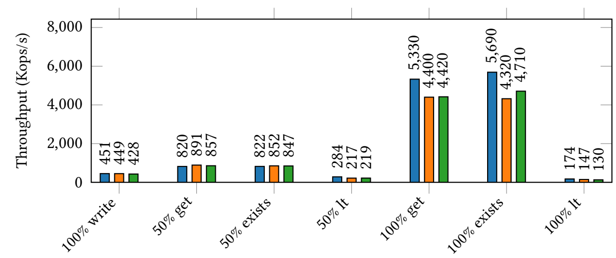
Figure 7: Throughput comparison for small values (64 bytes). Top: homogeneous access distribution ($\theta = 0$). Bottom: skewed distribution ($\theta = 2$).

Read Performance. The gap narrows for read operations under homogeneous access patterns. For get operations at 100% reads (Figure 7a, rightmost bars), RocksDB holds a 1.4× advantage. This residual gap reflects LSM-trees' cache efficiency advantage: sorted runs pack many small entries into each cached block, whereas TIDEHUNTER's position-based lookups touch one cached page per lookup. Notably, TIDEHUNTER's exists operation (Figure 7a) holds a 1.3× advantage over RocksDB at 100% reads, showing that our existence-check optimization provides value even for small entries.

Effect of Skew. Skewed workloads (Figure 7b) reveal an important exception to LSM-trees' small-value advantage. Under read-only workloads with high skew, TIDEHUNTER surpasses RocksDB by



(a) Homogeneous Distribution ($\theta = 0$)



(b) Skewed Distribution ($\theta = 2$)

Figure 8: Throughput comparison for small values (128 bytes). The 50% read cases include 50% writes. Top: homogeneous access distribution ($\theta = 0$). Bottom: skewed distribution ($\theta = 2$).

14% on get and 19% on exists. This reversal occurs because TIDEHUNTER's Large Table efficiently caches hot entries, and the WAL's append-only structure means recently written (and frequently accessed under skew) entries are contiguous, improving cache locality.

Summary. For small 64-byte values with homogeneous access patterns, LSM-tree-based systems retain a 1.4–2.2× advantage over TIDEHUNTER (1.4× on 100% get reads, 2.2× on writes), and TIDEHUNTER slightly leads on existence checks (1.3×). However, TIDEHUNTER regains the advantage under skewed read workloads where caching effects dominate. This suggests that TIDEHUNTER is better suited for workloads with larger values or significant temporal locality, while LSM-trees remain preferable for small-value workloads with homogeneous access patterns.

The results for 128-byte values are in Figure 8. The 128-byte value size represents a transitional point between small and large value performance characteristics, where neither architecture holds a decisive advantage across all workloads.

Write Performance. Under write-only workloads (Figure 8a, leftmost bars), RocksDB leads modestly with TIDEHUNTER and BlobDB both within 15%. At 128 bytes, TIDEHUNTER closes most of the write-throughput gap with LSM-trees: its single-write advantage nearly offsets LSM-trees' efficient small-entry handling, though RocksDB retains a modest lead.

Read Performance. For homogeneous read workloads, LSM-trees retain a modest advantage for value-fetching operations. At 100% reads, RocksDB holds a 1.2× advantage on get operations (Figure 8a).

However, TIDEHUNTER excels at existence checks: at a 50/50 read-write ratio, TIDEHUNTER outperforms RocksDB by 1.7× on exists operations (Figure 8a). This demonstrates that TIDEHUNTER's optimized existence check provides substantial benefits.

For reverse iterator operations (Figure 8a), TIDEHUNTER matches or beats LSM-trees in both write-heavy (by 30%) and read-heavy (by 21%) settings at 128 bytes; the iteration cost amortizes the per-record metadata across consecutive entries.

Effect of Skew. Under skewed access patterns (Figure 8b), TIDEHUNTER's in-memory caching provides a consistent advantage for read-heavy workloads. TIDEHUNTER surpasses RocksDB by 21% on get and 32% on exists at 100% reads with skew.

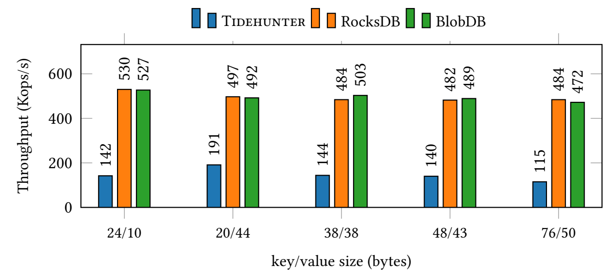
Summary. At 128 bytes, TIDEHUNTER and LSM-tree systems show more balanced performance compared to smaller values. TIDEHUNTER closes most of the write gap with LSM-trees, decisively outperforms them on existence checks (1.7× at 50/50), and matches or beats them on reverse iteration. LSM-trees retain only a narrow advantage (1.2×) for homogeneous 100% get operations. This transitional behavior indicates that TIDEHUNTER's advantages grow progressively as value sizes increase beyond this threshold.

6.2.3 Application-Workload Regimes. The previous subsections vary value size with a fixed 32-byte key. To characterize TIDEHUNTER's boundary on workloads with different shapes, we evaluate five key/value combinations drawn from production-trace studies of small-record key-value workloads at hyperscalers [54]. The combinations are (key/value, in bytes): (24/10) corresponding to RTDATA, (48/43) corresponding to ZippyDB, and three intermediate points (20/44), (38/38), and (76/50). These regimes have small entries dominated by key bytes, the opposite of TIDEHUNTER's favored large-value regime. They probe the small-record edge of the design rather than its core target. We pre-fill 500 GB of raw user bytes per backend and run a 30-minute measurement phase with 50% reads and 50% writes, under both homogeneous ($\theta = 0$) and skewed ($\theta = 2$) access. Results are in Figure 9.

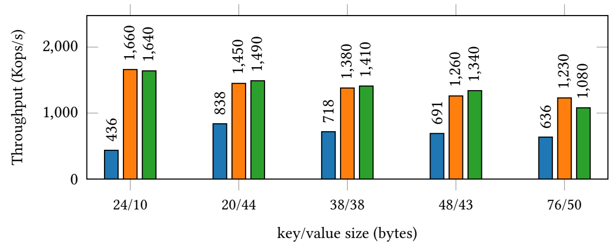
TIDEHUNTER regresses on small-record mixed workloads. Across all five combinations under homogeneous access (Figure 9a), TIDEHUNTER's throughput sits 2–4× below RocksDB and BlobDB. TIDEHUNTER's per-record WAL and index metadata are constant-size and therefore dominate when the entry itself is only 30–90 bytes; LSM-trees pack many such entries into each cached block and amortize their metadata cost over them.

The regression persists under skew on the 50/50 mix. Unlike the 64-byte case where skew lets TIDEHUNTER overtake LSM-trees for read-only get operations (Figure 7b), at these regimes RocksDB and BlobDB remain ahead under $\theta = 2$ across all five combinations (Figure 9b). Skew helps every backend, but the LSM block cache holds far more hot small records than TIDEHUNTER's Large Table at this scale, and the existing 50% writes prevent skew from collapsing into a pure cache-hit microbenchmark.

Summary. These results delineate TIDEHUNTER's deployment boundary. TIDEHUNTER is engineered for workloads where values are large enough that per-record metadata is amortized cheaply and where LSM-tree compaction is the dominant cost (content-addressable storage, deduplication, and the blockchain-validator workload that motivated its design). Workloads with key-dominated small records, such as those characterized by [54], fall outside this regime; for those, LSM-tree designs remain the appropriate choice.



(a) Homogeneous Distribution ($\theta = 0$)



(b) Skewed Distribution ($\theta = 2$)

Figure 9: Throughput on five small-record application workloads with 50% reads / 50% writes. K/V combinations are listed as "key/value" bytes, ordered by total entry size. (24/10) and (48/43) match the RTDATA and ZippyDB workload signatures from [54]; the other three are intermediate points. Top: homogeneous ($\theta = 0$). Bottom: skewed ($\theta = 2$).

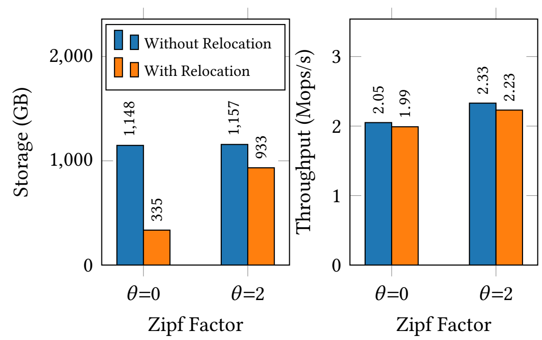


Figure 10: Effect of relocation on storage usage and throughput. Left: storage after mixed phase. Right: throughput during mixed phase. Relocation significantly reduces storage with minimal throughput impact.

Effect of Relocation. To measure relocation's storage and performance impact, we ran a 100%-delete workload under uniform (θ=0) and skewed (θ=2) access (Figure 10) on a 1 TB pre-fill of 1 KB values. Without relocation, deleted entries leave gaps that accumulate to roughly the original 1 TB under both patterns. With relocation, storage drops by 71% (uniform) and 19% (skewed); the smaller savings under skew reflect that only frequently accessed keys are deleted, leaving fewer reclaimable gaps. Throughput overhead is minimal: 3% uniform, 4% skewed.

6.2.4 *Write Amplification and Tail Latency Under Sustained Churn.*

We evaluate TIDEHUNTER's foreground behavior under workloads that continuously update or delete keys, asking: (i) what write amplification (WA) does the application observe at steady state, (ii) how much does enabling relocation cost in foreground throughput and tail latency, and (iii) can the relocation reclaim threshold be tuned without imposing a foreground tax?

Each configuration pre-fills 500 GB of (32 B key, 1 KB value) data with pure inserts, then runs a 60-minute pure-write measurement phase. Measurement operations split into overwrites, deletes, and fresh inserts; the mix is controlled by the (overwrite, delete) ratio. We compute WA as bytes written to the data device divided by application-visible bytes during measurement.

Strategy comparison (Table 2). We swept three relocation strategies (None, WalBased, IndexBased) across three workload mixes. Within each workload, WA is invariant across strategies to within ±0.02 and foreground throughput differs by less than 2%. Enabling relocation does not amplify writes: the rewrite cost runs on background threads with no read-path coupling and does not appear in the application's I/O accounting. The absolute WA of ≈2.0 in the 100% delete row is not caused by relocation (the no-relocation baseline shows the same value). Instead, it reflects the fixed per-record WAL metadata overhead: each delete writes only a ≈32 B tombstone as application data, so the constant metadata bytes are large relative to that payload, roughly doubling the ratio.

Cross-system contrast (Table 2, BlobDB rows). We rerun this sweep with BlobDB as a cross-system reference point on the same hardware, workload, and resource budget. BlobDB's compaction-driven value-log GC pays for sustained churn on every axis. Foreground throughput drops to 42–67% of TIDEHUNTER's across the three workloads. Write-path p99.9 latency inflates 7–43×, because of compaction threads sharing the foreground I/O path: most operations land cheaply in the memtable, while periodic compactions stall a small fraction of operations for milliseconds. WA is also higher in every workload: roughly 30% higher on overwrite-heavy mixes and nearly 3× on pure deletes. Each BlobDB delete propagates a tombstone through every LSM level via compaction and the blob file holding the deleted value must eventually be rewritten to drop the dead entry, cascading writes that dominate disk I/O even though the application's per-delete intent is only a 32 B key. TIDEHUNTER instead appends a 32 B tombstone and updates the in-memory index; the relocator is decoupled from the foreground.

Threshold sweep (Table 3). We sweep the relocation percentage (the bound on the fraction of WAL a single relocation pass may rewrite) over $\{1, 5, 10, 25, 50\}\%$ on the 50/50 mixed workload with WalBased relocation. WA stays flat at 1.19–1.20, throughput within 2%, and tail latency at p99 and p999 within 6% across the entire range. Operators can dial reclaim aggressiveness without paying a foreground tax.

Table 2: Foreground behavior under sustained measurement-phase churn. *WA*: bytes written to data device divided by application bytes written, measurement phase only. *Tput*: foreground operations per second. *p99/p999*: write-path latency in μs.

Workload	Strategy	WA	Tput	p99/p999
100% overwrite	None	1.14	857 K	62/76
	WalBased	1.15	853 K	61/76
	IndexBased	1.16	843 K	60/76
	BlobDB	1.54	361 K	2,190/3,279
50/50 mix	None	1.19	992 K	89/113
	WalBased	1.20	998 K	88/111
	IndexBased	1.22	992 K	88/112
	BlobDB	1.51	662 K	78/2,184
100% delete	None	2.12	2,070 K	115/157
	WalBased	1.98	2,080 K	116/159
	IndexBased	2.00	2,100 K	116/160
	BlobDB	5.82	959 K	51/1,104

Table 3: Threshold sweep on WalBased + 50/50 mixed workload. Columns as in Table 2.

Reclaim	WA	Tput	p99/p999
1%	1.19	995 K	88/111
5%	1.20	998 K	88/111
10%	1.19	1,000 K	88/111
25%	1.20	982 K	92/116
50%	1.20	999 K	88/111

6.2.5 *Recovery.*

We measure how long TIDEHUNTER takes to come back online after a restart, varying database size, snapshot cadence, and whether the shutdown was clean (Table 4). Each row corresponds to one restart on a pre-filled database; we report wall-clock recovery time and a first-read latency from 1,000 single-threaded random reads issued immediately after open.

Recovery time is essentially Value WAL replay. The other phases (acquiring the lock, reading the Control Region, opening the Value WAL, initializing the Large Table, opening the Index Store writer, and starting background threads) together take under 0.5% of the total in every run.

Cold-start scaling. Recovery time grows much more slowly than the database: a 10× size increase from 100 GB to 1 TB translates into only a 2.6× increase in recovery time. The amount of WAL replayed has a floor set by the snapshot interval, which is why the 100 GB and 500 GB rows replay roughly the same. Only the 1 TB row replays substantially more, because more writes had accumulated past the most recent snapshot. First-read p99 latency rises sharply with database size: the larger on-disk Index Store means each random read touches more cold pages.

Worst-case snapshot cadence. The biggest recovery lever is turning periodic snapshots on at all: a 1 TB database without snapshots must replay the entire WAL (nearly 30 min), while any finite cadence cuts this by 3–5×. Within finite cadences, smaller is not

Table 4: Recovery measurements. "Repl." shows bytes processed during recovery. "Total" is wall-clock recovery time. "First-read" percentiles are over 1,000 random keys read. In the snapshot-interval block, ∞ disables periodic snapshots.

	DB (GB)	Snap (GB)	Repl. (GB)	Total (s)	First-read (μs) p50/p99
Cold-start vs DB size, 128 GB snapshot					
cold-100	100	128	109	180	4/7
cold-500	500	128	140	245	81/350
cold-1tb	1,024	128	245	465	352/995
Snapshot interval at 1 TB					
snap-16	1,024	16	263	432	299/945
snap-64	1,024	64	243	401	312/922
snap-128	1,024	128	207	350	289/985
snap-256	1,024	256	317	566	307/977
snap-∞	1,024	–	1,116	1,791	127/1,328
200 GB fill, crash during relocation					
crash	200	128	85	146	82/232

always faster, because a TIDEHUNTER snapshot writes only a Control Region pointer; it does not synchronously flush dirty cells from the Large Table. That pointer can advance no further than the WAL offset of the oldest dirty cell, so under sustained writes the recovery start point lags the WAL head by much more than the snapshot interval. The practical takeaway: enable periodic snapshots; tuning the cadence yields only modest additional gains.

Crash during relocation. We also checked that an unclean shutdown during relocation leaves the database no worse than a clean one. We pre-filled 200 GB, ran continuous Value WAL relocation in the background, and killed the process abruptly: no destructors ran and no final fsync was issued, so only data already in the OS page cache survived. Recovery completes promptly (Table 4); first-read latency is in line with cold-start results, showing that continuous relocation does not degrade post-crash read performance.

6.2.6 Memory Footprint and Sensitivity. We quantify TIDEHUNTER's RAM usage along three axes: an analytical breakdown of per-key and per-cell metadata, a runtime instrumentation of the main eval point that measures the cache layer's actual hit ratio and eviction-rate behavior, and sensitivity sweeps over the configuration parameters that govern the memory–performance trade-off.

Per-key and per-cell footprint. The Large Table stores each key as a fixed-format entry in its cell's sorted in-memory index, which uses the same layout as the on-disk index (§4): key size +12 B per entry (8 B Value WAL offset + 4 B length word), so 32 B keys cost 44 B per resident entry. Per cell, the always-resident metadata is approximately 440 B: a 400 B per-cell metadata block (cell identifier, a handle to the on-disk index, the pending-update table, four metric-delta trackers, optional Bloom-filter and value-cache slots) plus a 40 B row-level wrapper holding the cell mutex and the entries array. With the default 2^{17} cells, cell metadata totals about 55 MiB. Dirty-but-unflushed entries are buffered in a per-cell write buffer whose size is bounded by the per-cell dirty-key cap (`max_dirty_keys`), which defaults to 1024 entries per cell; at ~80 B per entry (32 B key + 16 B value + BTreeMap node overhead), this buffer is bounded at ~80 KiB per cell.

Cache layer and measured hit ratio. TIDEHUNTER's only configured value cache is the memory-mapped Value WAL (128 GiB window on a 256 GB machine, with physical residency managed by the OS page cache); the optional LRU is disabled in all paper experiments. We instrumented TIDEHUNTER's runtime metrics on the main eval point (1 TB pre-fill, 32 B keys, 1 KB values, 50/50 mixed Get, $\theta = 0$, 30 min).

Of ~600 M lookups, 4.3% resolved in the Large Table in-memory index and 95.7% required the on-disk Index Store; the LRU and Bloom sources accounted for 0%, because the LRU is disabled, and our read workload follows YCSB's positive-lookup convention (§6.1). Under this uniform workload over $\sim 10^9$ keys, the Large Table's role is not to serve reads directly but to (a) write-batch new keys until the per-cell dirty-key cap is exceeded and the cell is flushed to the Index Store, and (b) hold the position estimator that resolves on-disk lookups in ≤ 2 disk reads (§6.3).

Eviction dynamics. The measurement phase issued 694 K flushes (~370/s), which flushed 7.1 B keys and churned 313 GB of index over 30 minutes (~170 MB/s of Index Store writes). Flush backpressure (§5) triggered ~3 M times, the dominant source of the foreground throughput variance in Table 1, and the snapshot force-relocated 7.3 K cells whose on-disk index lay below the relocation cutoff. At the end of the phase, TIDEHUNTER held 137.4 GB of mmap'd WAL and 6.44 GB of estimated total memory: 4.0 GB of resident key bytes, 3.7 GB of sorted in-memory index, and 82 M dirty keys awaiting flush. These in-memory totals oscillate between 1.4 GB and 6.2 GB, since the end-of-phase sample can land anywhere in the flush–accumulate cycle; the mean working set is ≈4 GB.

Sensitivity to Bloom-filter FPR. We swept the Bloom target false-positive rate p over four points from 0.001 to 0.1, at a fixed workload (1 KB values, 32 B keys, 100% Get, $\theta = 0$, 1 TB pre-fill). Because our read workload follows YCSB's positive-lookup convention (§6.1), every Get queries a key that is present in the database, so the Bloom always returns a true positive regardless of p. Foreground throughput is therefore essentially flat across the sweep (456–470 K ops/s, <3% spread), so for this workload the parameter trades only memory. The Bloom array size per cell is $(\lceil -n \ln p / (\ln 2)^2 \rceil)$ bits, where n is the per-cell Bloom capacity in keys (`bloom_filter_count`). With the default $n = 8192$, this falls from 14.4 bits per key at $p = 0.001$ to 4.8 at $p = 0.1$. Summed over the default 2^{17} cells, the total Bloom footprint at $p = 0.01$ is ~1.3 GiB.

Sensitivity to cell sizing and dirty-key thresholds. We swept the two memory-shaping parameters in Table 5: the cell count (`num_mutexes`, with one cell per mutex) and the per-cell dirty-key cap (`max_dirty_keys`), on the standard 50/50 / $\theta = 0$ / 1 KB workload. The default 2^{17} cells and 1024-entry dirty cap are at the throughput sweet spot. On the cell-count axis, throughput drops relative to the 2^{17} default by 30% at 2^{14} (each cell holds ~8× more keys, so each merge of pending writes into the sorted in-memory index churns a larger array), by 4% at 2^{16}, by 5% at 2^{19}, and by 22% at 2^{20} (per-cell metadata pressure on cache lines outweighs the reduced per-cell working set). On the dirty-key axis: dropping to 64 entries/cell collapses throughput by 49% from premature unloads, and the same regime blows

Table 5: Memory-sensitivity sweeps.

Parameter	Value	Tput	p99/p999 (μs)
Total mmap window (across both WALs)			
window	32 GiB	640 K	201 / 285
	64 GiB	660 K	199 / 280
	128 GiB (default)	677 K	207 / 469
	256 GiB	612 K	311 / 896
Cell count (one mutex per cell)			
cells	2^{14} (16,384)	429 K	409 / 1,276
	2^{16} (65,536)	591 K	333 / 830
	2^{17} (131,072, default)	612 K	312 / 878
	2^{19} (524,288)	581 K	365 / 901
	2^{20} (1,048,576)	478 K	593 / 1,381
Per-cell dirty-key cap			
cap	64	312 K	653 / 14,422
	256	553 K	386 / 934
	1,024 (default)	612 K	312 / 878
	4,096	588 K	352 / 1,012
	16,384	603 K	324 / 934

out the p99.9 to 14 ms (versus ~900 μs at the default); raising to 4096 or 16384 yields no measurable gain (the working set's resident footprint plateaus once cells flush near their natural cadence). Tail latency on the cell-count axis moves with throughput in the same direction: the 2^{17} default minimizes both, while the 2^{14} and 2^{20} extremes raise p99.9 to 1.3–1.4 ms. Both parameter axes validate the production defaults.

Sensitivity to mmap window size. Sweeping the per-WAL fragment count (`max_maps`) at a fixed 1 GiB fragment size traces out the mmap-window curve in Table 5. Throughput rises from 640 K ops/s at a 32 GiB total mmap window to a peak of 677 K at 128 GiB (64 fragments per WAL), then drops to 612 K at 256 GiB (128 fragments per WAL). The drop at the largest budget reflects TIDEHUNTER's mmap address space approaching the machine's 256 GB of physical RAM: with essentially the whole RAM claimed as mmap'd address space, the kernel must page WAL fragments in and out and has no headroom for write buffers, OS metadata, or non-WAL state. Halving the configured window to 128 GiB gives the OS room to manage residency efficiently while still covering the working set under uniform 50/50 read traffic. Tail latency tracks the same curve: p99 stays at ~200 μs across the 32–128 GiB range and climbs to 311 μs at 256 GiB; p99.9 stays at 280–290 μs at the two smallest windows, rises to 469 μs at the 128 GiB throughput peak, and reaches 896 μs at the RAM-saturated 256 GiB point. The 128 GiB setting (64 fragments per WAL, applied to both the Value WAL and the Index Store) is therefore both the production configuration and the throughput-optimal choice on this hardware, and we use it throughout the paper.

6.3 Index Format Comparison

We microbenchmark the lookup performance of TIDEHUNTER's two persistent index formats: the optimistic index and a naive baseline.

- **Header-based index**: This format uses a fixed-size header of 128 entries (1 KB total) to partition keys into micro-cells based on their prefix. Each header entry stores offsets pointing to a contiguous region of sorted key-value pairs. A lookup requires exactly two I/O operations: reading the 8-byte header entry to determine the data region, followed by reading and binary searching the data region.
- **Optimistic index** (§4.2): This format assumes a uniform key distribution to estimate the probable location of a key within the sorted index file. It uses a window-based search strategy with a default window size of 800 entries. The search begins at the estimated offset and iteratively moves the window until the target key falls within bounds, then performs a binary search. This approach eliminates header overhead and can achieve single-roundtrip lookups when the distribution estimate is accurate.

Data Generation. We generate 25 000 separate index files, each containing 1 000 000 randomly populated entries. Each entry consists of a 32-byte key generated uniformly at random and an 8-byte WAL position. The indices are serialized to disk using their respective formats, resulting in approximately 1 TB of data per format.

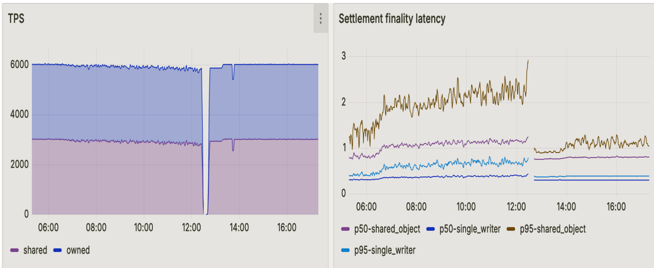
Benchmark Configuration. The benchmark performs 10 000 000 lookups. Lookups are grouped into batches of 1 000 operations for timing measurements. The window size for the optimistic index is set to 800 entries. We evaluate both single- and multi-threaded configurations, and optionally enable direct I/O to bypass the operating system's page cache.

Workload. The benchmark uses a random access pattern where each lookup selects a uniformly random index from the 25 000 available indices and queries a randomly generated 32-byte key. Since the lookup keys are generated independently of the keys stored in the indices, this workload consists predominantly of negative lookups (keys not present in the index), which represents a worst-case scenario for the optimistic index as it may require multiple window movements to determine key absence.

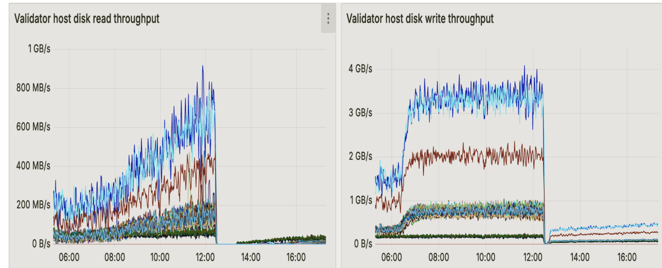
Results. Figure 12 shows the lookup throughput of both index formats as a function of window size, at 1, 16, and 48 threads. The optimistic index's optimal window shrinks as concurrency rises, from 1600 at low thread counts to 200–400 at high thread counts, while the header-based index is insensitive to window size.

With direct I/O enabled, the optimistic index outperforms the header-based index at all thread counts, achieving up to 194K lookups/s at 48 threads compared to 156K for the header-based index (a 24% improvement). Without direct I/O, both indices plateau at high thread counts due to page cache contention, with the optimistic index achieving 71K lookups/s compared to 57K for the header-based index.

Figure 12 shows the sensitivity of the optimistic index to window size across different thread counts. The header-based index maintains constant throughput regardless of window size (since it always performs exactly two I/O operations per lookup), while the optimistic index exhibits a clear trade-off: smaller windows require more iterations but read less data per iteration, while larger windows reduce iterations but increase I/O volume. The optimal window size decreases as thread count increases, reflecting the benefits of smaller I/O operations under high concurrency.



(a) Transaction latency. RocksDB (before 12:30) degrades over time; TIDEHUNTER (after 12:30) stays stable.



(b) Disk I/O. TIDEHUNTER (after 12:30) achieves significantly lower disk read and write rates.

Figure 11: Sui blockchain under 6,000 TPS load: the same 150 validators run RocksDB before 12:30 and TIDEHUNTER after.

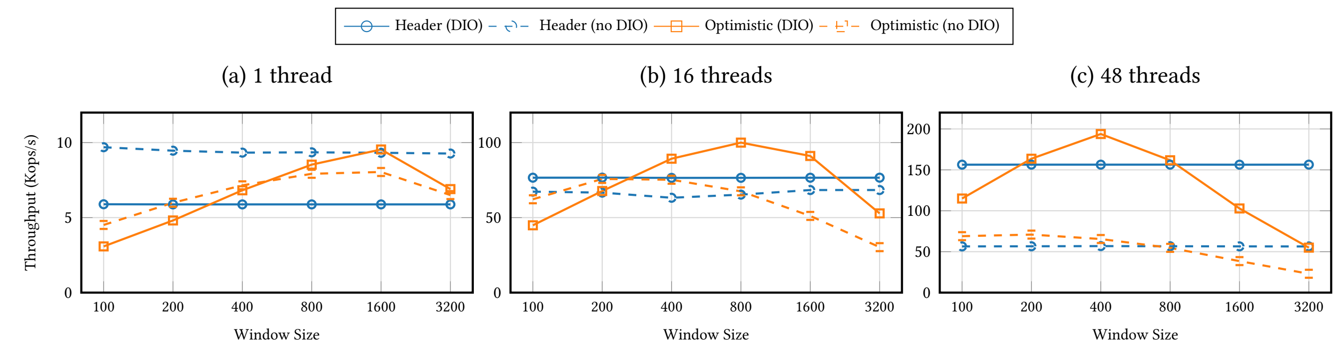


Figure 12: Index lookup throughput vs. window size for different thread counts. The header-based index maintains constant throughput (flat lines), while the optimistic index performance varies with window size. The optimal window size decreases as concurrency increases.

6.4 Case Study: Sui Blockchain

To demonstrate real-world performance, we integrated TIDEHUNTER into the Sui blockchain, which currently uses RocksDB as its database backend. We replaced Sui's validator storage layer with TIDEHUNTER and deployed in a cluster of 150 geographically distributed machines under an artificial load of 6,000 transactions per second.

The hosts started the experiment using RocksDB and switched to using TIDEHUNTER. These results are presented in Figures 11a and 11b. The first thing to notice is that under the sustained load of 6,000 TPS, RocksDB performance degrades after several hours, resulting in significant latency increases (Figure 11a before 12:30). In contrast, once TIDEHUNTER is deployed the same hosts maintain stable latency throughout the test period (Figure 11a, after 12:30). The same characteristics are also present in the disk I/O utilization (Figure 11b), where the high write amplification of RocksDB results in 1–4 GB/s write throughput demand, whereas the need to search deep inside the LSM-Tree results in 0.2–0.8 GB/s read throughput demand. In contrast, once TIDEHUNTER is enabled the demand of the disk resource drops to less than 100 MB/s.

Since TIDEHUNTER sustained 6,000 TPS without any issue, we ran a second experiment with a load of 8,500 TPS. The performance of the same cluster using TIDEHUNTER under this increased load is demonstrated in Figure 13, where both throughput and latency remain steady during five hours of sustained load.

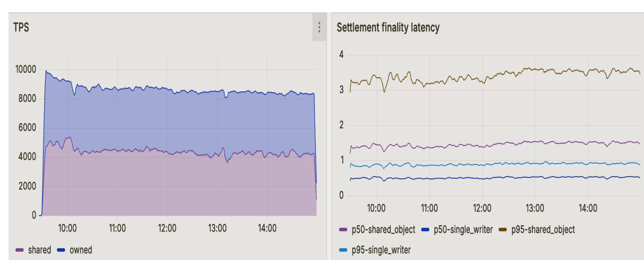


Figure 13: Performance of Sui using TIDEHUNTER under 8500 TPS load. Metrics remain steady, unlike RocksDB with 6000 TPS (Figure 11a, before 12:30).

As of January 2026, TIDEHUNTER is deployed on Sui's devnet, and several testnet full nodes have been permanently migrated from RocksDB to TIDEHUNTER.

7 Related Work

LSM-Tree Stores. LSM-trees [50] organize data into sorted runs across multiple levels, trading write amplification for read performance through background compaction. Stores like RocksDB [11, 26], LevelDB [22], and PebblesDB [56] have become industry standards, with various optimizations including lazy leveling [20], optimal Bloom filter allocation [19], and concurrent compaction [27]. Despite these improvements, compaction fundamentally rewrites data 5–10× across levels. TIDEHUNTER sidesteps this entirely: its append-only WAL eliminates multi-level compaction, achieving 6.8× higher write throughput than RocksDB.

Key-Value Separation. WiscKey [41] pioneered storing values in a separate log (vLog), dramatically reducing write amplification for large values. Production successors include BlobDB [42], Titan [55], DiffKV [40], HashKV [12], BVLSM [39], Kreon [53], and BadgerDB [24], which variously optimize value placement, garbage collection, and flash-friendliness. Most keep an LSM-tree for keys, inheriting its compaction overhead: key compaction still causes write stalls, multi-level lookups still suffer under uniform key distributions, and vLog garbage collection contends with foreground writes. TIDEHUNTER instead makes the WAL the unified value store with lazy-flushed index tables, avoiding both LSM compaction and separate vLog management.

B-Tree and Memory-Mapped Stores. LMDB [15] uses copy-on-write B+ trees with shadow paging, enabling zero-copy reads but limited to single-writer concurrency. CedrusDB [63] combines memory-mapped lazy-tries with a WAL and is the most architecturally similar prior work to TIDEHUNTER. However, CedrusDB's in-place trie updates require per-node locking during traversal, limiting write concurrency to approximately eight threads; its read path must walk the trie from root to leaf regardless of key distribution. TIDEHUNTER's append-only WAL with atomic position allocation sustains high concurrent write throughput, and its optimistic index bypasses tree traversal entirely for on-disk lookups. Traditional B-tree systems with logging and locking include BerkeleyDB [49] and redb [8]. Unlike these, TIDEHUNTER's append-only design avoids page splits and tree rebalancing entirely.

Hardware-Optimized Stores. KVell [36] maximizes NVMe bandwidth via per-CPU slices without data sorting, sharing TIDEHUNTER's avoidance of sort overhead. SpanDB [14] places WAL on fast NVMe with SPDK; FASTER [13] uses epoch-protected hybrid logs. SILK [5] and TRIAD [4] address LSM write stalls via I/O scheduling and tiered storage. TIDEHUNTER instead eliminates LSM compaction architecturally, using only standard mmap and O_DIRECT interfaces.

Log-as-Database Systems. Several systems share TIDEHUNTER's philosophy of treating the log as the primary data store, but at different abstraction levels and for different bottlenecks. Amazon Aurora [61] sends only redo log records over the network; its storage tier materializes pages from redo records, reducing *network* amplification in a distributed RDBMS. ERMIA [34] replaces in-memory heap pages with log-structured version records and OID-array indirection, targeting MVCC efficiency in DRAM. LLAMA [38] provides a latch-free, log-structured cache/storage substrate for access methods like the Bw-Tree [37], using delta chains and CAS-based page updates; it relies on cleaning and consolidation both to reclaim space and to shorten delta chains, which also improves read performance. In contrast, TIDEHUNTER targets *disk* write amplification for large-value key-value workloads: its WAL permanently stores values (not redo records or version deltas), and its relocation is optional because read performance is independent of WAL fragmentation.

Update-in-Place and Learned Indexes. TreeLine [64] also rejects LSM compaction on modern NVMe SSDs, but uses update-in-place B-tree pages with record caching and model-guided page grouping, embracing random writes where SSDs' sequential-to-random gap has narrowed. TIDEHUNTER takes the opposite approach: strictly sequential, append-only writes that achieve minimal write amplification while trading away sorted iteration. Kraska et al. [35] observe that indexes are models of the data distribution's CDF and show that learned models (from linear regression to neural networks) can replace traditional index structures. TIDEHUNTER's optimistic index shares this CDF-as-position-estimator insight but exploits the uniform key distribution directly, where the CDF is a trivial linear function, requiring no training or model maintenance and naturally supporting dynamic data.

Learned and Adaptive Indexes. Machine learning techniques can replace traditional index structures with models that predict key positions. Bourbon [18] adds learned indexes atop WiscKey using piecewise linear regression for 1.23–1.78× faster lookups. Kraska et al. [35] propose replacing B-trees and Bloom filters with neural networks, trading training overhead for lookup speed. ADOC [65] automatically tunes RocksDB parameters, achieving 87.9% write stall reduction. TIDEHUNTER uses conventional Bloom filters and hash-based indexing for predictability, prioritizing write performance over learned lookup optimizations.

Alternative Tree Structures. SplinterDB [16] uses size-tiered Bε-trees, achieving 6–10× faster insertions than RocksDB by batching updates in nodes. TerarkDB [10] uses compressed searchable SSTs optimized for reduced tail latency. ForestDB [1] uses HB+-tries for document workloads, while the Bw-Tree [37] achieves lock-freedom through delta records and epoch-based reclamation. TIDEHUNTER's append-only WAL avoids the complexity of tree maintenance entirely, trading sorted iteration for minimal write amplification.

Rust Implementations. Rust's memory safety guarantees make it increasingly popular for storage engines. Fjall [28] provides modern LSM-trees with built-in KV separation. AgateDB [59] ports BadgerDB's WiscKey-style design for TiKV. sled [48] uses flash-optimized Bw-trees with lock-free reads. TIDEHUNTER shares Rust's safety benefits while introducing a novel WAL-centric architecture distinct from these LSM and B-tree based designs.

Blockchain Storage. Blockchain systems require high-throughput persistent storage with strict durability. TIDEHUNTER was developed for Sui [9], where transaction objects (typically 1KB–1MB) drive the large-value workload. Aptos uses JellyfishMerkle [29] sparse Merkle trees for state management, while Ethereum [62] evolved from LevelDB to Pebble for state storage [30]. TIDEHUNTER could serve as the underlying storage for such Merkle tree implementations.

Write-Ahead Logging. ARIES [46] established foundational WAL protocols with steal and no-force buffer management for B-tree databases. Silo [60] uses epoch-based transaction processing with group commit, which inspired TIDEHUNTER's batched WAL syncing.

Hekaton [25] optimizes logging for in-memory OLTP, while Write-Behind Logging [3] targets persistent memory. Unlike traditional systems where WAL is auxiliary for recovery, TIDEHUNTER's WAL is the primary data store with lazy index flushing.

8 Conclusion

TIDEHUNTER is a key-value store that eliminates value compaction by treating the WAL as permanent storage: values are written once and relocated only rarely, to reclaim space, while small, lazily-flushed index tables map keys to WAL positions. On 1 KB values, TIDEHUNTER achieves 6.8× higher write throughput than RocksDB, 2.0× faster point queries, and 21.1× faster existence checks; LSM-trees regain only a narrow 1.2–1.4× advantage on homogeneous point queries below 128 bytes. Sui validator integration confirms these gains hold in production, with stable performance under loads that collapse RocksDB-backed systems. The tradeoff is generality for efficiency on an increasingly common pattern: hash-keyed, kilobyte-scale, write-heavy workloads in content-addressable storage, deduplication, and blockchain systems.

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References

[1] Jung-Sang Ahn, Chiyoung Seo, Ravi Mayuram, Rahim Yaseen, Jin-Soo Kim, and Seungryoul Maeng. 2016. ForestDB: A Fast Key-Value Storage System for Variable-Length String Keys. *IEEE Trans. Comput.* 65, 3 (2016), 902–915.

[2] David G. Andersen, Jason Franklin, Michael Kaminsky, Amar Phanishayee, Lawrence Tan, and Vijay Vasudevan. 2009. FAWN: A Fast Array of Wimpy Nodes. In *SOSP*. ACM, Big Sky, MT, 1–14.

[3] Joy Arulraj, Matthew Perron, and Andrew Pavlo. 2016. Write-Behind Logging. *Proceedings of the VLDB Endowment* 10, 4 (2016), 337–348.

[4] Oana Balmau, Diego Didona, Rachid Guerraoui, Willy Zwaenepoel, Huapeng Yuan, Aashray Arora, Karan Gupta, and Pavan Konka. 2017. TRIAD: Creating Synergies Between Memory, Disk and Log in Log Structured Key-Value Stores. In *USENIX ATC*. USENIX Association, Santa Clara, CA, 363–375.

[5] Oana Balmau, Florin Dinu, Willy Zwaenepoel, Karan Gupta, Ravishankar Chandhiramoorthi, and Diego Didona. 2019. SILK: Preventing Latency Spikes in Log-Structured Merge Key-Value Stores. In *USENIX ATC*. USENIX Association, Renton, WA, 753–766.

[6] Doug Beaver, Sanjeev Kumar, Harry C. Li, Jason Sobel, and Peter Vajgel. 2010. Finding a Needle in Haystack: Facebook's Photo Storage. In *OSDI*. USENIX Association, Vancouver, Canada, 47–60.

[7] Juan Benet. 2014. IPFS - Content Addressed, Versioned, P2P File System. arXiv:1407.3561.

[8] Christopher Berner. 2023. redb: An Embedded Key-Value Database in Pure Rust. Retrieved July 27, 2026 from https://github.com/cberner/redb

[9] Sam Blackshear, Andrey Chursin, George Danezis, Anastasios Kichidis, Lefteris Kokoris-Kogias, Xun Li, Mark Logan, Ashok Menon, Todd Nowacki, Alberto Sonnino, Brandon Williams, and Lu Zhang. 2024. Sui Lutris: A Blockchain Combining Broadcast and Consensus. In *CCS*. ACM, Salt Lake City, UT, 2606–2620. https://doi.org/10.1145/3658644.3670286

[10] ByteDance. 2021. TerarkDB: A RocksDB-Compatible Key-Value Store With Searchable Compression. Retrieved July 27, 2026 from https://github.com/bytedance/terarkdb

[11] Zhichao Cao, Siying Dong, Sagar Vemuri, and David H.C. Du. 2020. Characterizing, Modeling, and Benchmarking RocksDB Key-Value Workloads at Facebook. In *FAST*. USENIX Association, Santa Clara, CA, 209–223.

[12] Helen H. W. Chan, Yongkun Li, Patrick P. C. Lee, and Yinlong Xu. 2018. HashKV: Enabling Efficient Updates in KV Storage via Hashing. In *USENIX ATC*. USENIX Association, Boston, MA, 1007–1019.

[13] Badrish Chandramouli, Guna Prasaad, Donald Kossmann, Justin Levandoski, James Hunter, and Mike Barnett. 2018. FASTER: A Concurrent Key-Value Store with In-Place Updates. In *SIGMOD*. ACM, Houston, TX, 275–290.

[14] Hao Chen, Chaoyi Ruan, Cheng Li, Xiaosong Ma, and Yinlong Xu. 2021. SpanDB: A Fast, Cost-Effective LSM-tree Based KV Store on Hybrid Storage. In *FAST*. USENIX Association, Virtual, 17–32.

[15] Howard Chu. 2011. MDB: A Memory-Mapped Database and Backend for OpenLDAP. In *LDAPCon*. LDAPCon, Heidelberg, Germany, 1–10.

[16] Alexander Conway, Abhishek Gupta, Vijay Chidambaram, Martin Farach-Colton, Richard Spillane, Amy Tai, and Rob Johnson. 2020. SplinterDB: Closing the Bandwidth Gap for NVMe Key-Value Stores. In *USENIX ATC*. USENIX Association, Virtual, 49–63.

[17] Brian F. Cooper, Adam Silberstein, Erwin Tam, Raghu Ramakrishnan, and Russell Sears. 2010. Benchmarking Cloud Serving Systems with YCSB. In *SoCC*. ACM, Indianapolis, IN, 143–154. https://doi.org/10.1145/1807128.1807152

[18] Yifan Dai, Yien Xu, Aishwarya Ganesan, Ramnatthan Alagappan, Brian Kroth, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. 2020. From WiscKey to Bourbon: A Learned Index for Log-Structured Merge Trees. In *OSDI*. USENIX Association, Virtual, 155–171.

[19] Niv Dayan, Manos Athanassoulis, and Stratos Idreos. 2017. Monkey: Optimal Navigable Key-Value Store. In *SIGMOD*. ACM, Chicago, IL, 79–94.

[20] Niv Dayan and Stratos Idreos. 2018. Dostoevsky: Better Space-Time Trade-Offs for LSM-Tree Based Key-Value Stores via Adaptive Removal of Superfluous Merging. In *SIGMOD*. ACM, Houston, TX, 505–520.

[21] Javier de la Rúa Martínez, Fabio Buso, Antonios Kouzoupis, Alexandru A. Ormenisan, Salman Niazi, Davit Bzhalava, Kenneth Mak, Victor Jouffrey, Mikael Ronström, Raymond Cunningham, Ralfs Zangis, Dhananjay Mukhedkar, Ayushman Khazanchi, Vladimir Vlassov, and Jim Dowling. 2024. The Hopsworks Feature Store for Machine Learning. In *SIGMOD Companion*. ACM, Santiago, Chile, 135–147.

[22] Jeffrey Dean and Sanjay Ghemawat. 2011. LevelDB. Retrieved July 27, 2026 from https://github.com/google/leveldb

[23] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT*. Association for Computational Linguistics, Minneapolis, MN, 4171–4186.

[24] Dgraph Labs. 2017. BadgerDB: Fast Key-Value DB in Go. Retrieved July 27, 2026 from https://github.com/dgraph-io/badger

[25] Cristian Diaconu, Craig Freedman, Erik Ismert, Per-Åke Larson, Pravin Mittal, Ryan Stonecipher, Nitin Verma, and Mike Zwilling. 2013. Hekaton: SQL Server's Memory-Optimized OLTP Engine. In *SIGMOD*. ACM, New York, NY, 1243–1254.

[26] Siying Dong, Andrew Kryczka, Yanqin Jin, and Michael Stumm. 2021. RocksDB: Evolution of Development Priorities in a Key-Value Store Serving Large-Scale Applications. *ACM Transactions on Storage* 17, 4 (2021), 1–32.

[27] Robert Escriva. 2013. HyperLevelDB: A Fork of LevelDB Optimized for Concurrency. Retrieved July 27, 2026 from https://github.com/rescrv/HyperLevelDB

[28] fjall-rs. 2025. Fjall: Log-Structured Embeddable Key-Value Storage Engine in Rust. Retrieved July 27, 2026 from https://github.com/fjall-rs/fjall

[29] Zhenhuan Gao, Yuxuan Hu, and Qinfan Wu. 2021. *Jellyfish Merkle Tree*. Technical Report. Diem Association.

[30] go-ethereum. 2023. Go Ethereum (Geth) v1.13.0 Release Notes. Retrieved July 27, 2026 from https://blog.ethereum.org/2023/09/12/geth-v1-13-0

[31] Fanglu Guo and Petros Efstathopoulos. 2011. Building a High-performance Deduplication System. In *USENIX ATC*. USENIX Association, Portland, OR, 285–298.

[32] Haochen He, Erci Xu, Shanshan Li, Zhouyang Jia, Si Zheng, Yue Yu, Jun Ma, and Xiangke Liao. 2023. When Database Meets New Storage Devices: Understanding and Exposing Performance Mismatches via Configurations. *Proceedings of the VLDB Endowment* 16, 7 (2023), 1712–1725.

[33] X.-Y. Hu, E. Eleftheriou, R. Haas, I. Iliadis, and R. Pletka. 2009. Write Amplification Analysis in Flash-Based Solid State Drives. In *SYSTOR*. ACM, Haifa, Israel, 1–9.

[34] Kangnyeon Kim, Tianzheng Wang, Ryan Johnson, and Ippokratis Pandis. 2016. ERMIA: Fast Memory-Optimized Database System for Heterogeneous Workloads. In *SIGMOD*. ACM, San Francisco, CA, 1675–1687.

[35] Tim Kraska, Alex Beutel, Ed H. Chi, Jeffrey Dean, and Neoklis Polyzotis. 2018. The Case for Learned Index Structures. In *SIGMOD*. ACM, Houston, TX, 489–504.

[36] Baptiste Lepers, Oana Balmau, Karan Gupta, and Willy Zwaenepoel. 2019. KVell: The Design and Implementation of a Fast Persistent Key-Value Store. In *SOSP*. ACM, Huntsville, ON, Canada, 447–461.

[37] Justin J. Levandoski, David B. Lomet, and Sudipta Sengupta. 2013. The Bw-Tree: A B-tree for New Hardware Platforms. In *ICDE*. IEEE, Brisbane, Australia, 302–313.

[38] Justin J. Levandoski, David B. Lomet, and Sudipta Sengupta. 2013. LLAMA: A Cache/Storage Subsystem for Modern Hardware. *Proceedings of the VLDB Endowment* 6, 10 (2013), 877–888.

[39] Ming Li, Wendi Cheng, Jiahe Wei, Xueqiang Shan, Weikai Liu, Xiaonan Zhao, and Xiao Zhang. 2025. BVLSM: Write-Efficient LSM-Tree Storage via WAL-Time Key-Value Separation. arXiv:2506.04678.

[40] Yongkun Li, Zhen Liu, Patrick P. C. Lee, Jiayu Wu, Yinlong Xu, Yi Wu, Liu Tang, Qi Liu, and Qiu Cui. 2021. Differentiated Key-Value Storage Management

for Balanced I/O Performance. In *USENIX ATC*. USENIX Association, Virtual, 673–687.

[41] Lanyue Lu, Thanumalayan Sankaranarayana Pillai, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. 2016. WiscKey: Separating Keys from Values in SSD-conscious Storage. In *FAST*. USENIX Association, Santa Clara, CA, 133–148.

[42] Meta. 2021. BlobDB: RocksDB's Key-Value Separation for Large Values. Retrieved July 27, 2026 from https://github.com/facebook/rocksdb/wiki/BlobDB

[43] Meta. 2021. Integrated BlobDB. Retrieved July 27, 2026 from https://rocksdb.org/blog/2021/05/26/integrated-blob-db.html

[44] Meta. 2024. RocksDB: A Persistent Key-Value Store for Flash and RAM Storage. Retrieved July 27, 2026 from https://github.com/facebook/rocksdb

[45] Meta. 2024. RocksDB: Setup Options and Basic Tuning. Retrieved July 27, 2026 from https://github.com/facebook/rocksdb/wiki/Setup-Options-and-Basic-Tuning

[46] C. Mohan, Don Haderle, Bruce Lindsay, Hamid Pirahesh, and Peter Schwarz. 1992. ARIES: A Transaction Recovery Method Supporting Fine-Granularity Locking and Partial Rollbacks Using Write-Ahead Logging. *ACM Transactions on Database Systems* 17, 1 (1992), 94–162.

[47] Mysten Labs. 2025. TideHunter: Source Code Repository. Retrieved July 27, 2026 from https://github.com/MystenLabs/tidehunter

[48] Tyler Neely. 2019. sled: The Champagne of Beta Embedded Databases. Retrieved July 27, 2026 from https://github.com/spacejam/sled

[49] Michael A. Olson, Keith Bostic, and Margo I. Seltzer. 1999. Berkeley DB. In *USENIX ATC*. USENIX Association, Monterey, CA, 183–191.

[50] Patrick O'Neil, Edward Cheng, Dieter Gawlick, and Elizabeth O'Neil. 1996. The Log-Structured Merge-Tree (LSM-Tree). *Acta Informatica* 33, 4 (1996), 351–385.

[51] OpenMetal. 2024. OpenMetal: On-Demand Private Clouds. Retrieved July 27, 2026 from https://openmetal.io

[52] Laurel Orr, Atindriyo Sanyal, Xiao Ling, Karan Goel, and Megan Leszczynski. 2021. Managing ML Pipelines: Feature Stores and the Coming Wave of Embedding Ecosystems. *Proceedings of the VLDB Endowment* 14, 12 (2021), 3178–3181.

[53] Anastasios Papagiannis, Giorgos Saloustros, Giorgos Xanthakis, Giorgos Kalaentzis, Pilar Gonzalez-Ferez, and Angelos Bilas. 2021. Kreon: An Efficient Memory-Mapped Key-Value Store for Flash Storage. *ACM Transactions on Storage* 17, 1 (2021), 7:1–7:32.

[54] Chanyoung Park, Jungho Lee, Chun-Yi Liu, Kyungtae Kang, Mahmut Taylan Kandemir, and Wonil Choi. 2025. AnyKey: A Key-Value SSD for All Workload Types. In *ASPLOS*. ACM, Rotterdam, Netherlands, 47–63. https://doi.org/10.1145/3669940.3707279

[55] PingCAP. 2019. Titan: A RocksDB Plugin for Key-Value Separation. Retrieved July 27, 2026 from https://github.com/tikv/titan

[56] Pandian Raju, Rohan Kadekodi, Vijay Chidambaram, and Ittai Abraham. 2017. PebblesDB: Building Key-Value Stores using Fragmented Log-Structured Merge Trees. In *SOSP*. ACM, Shanghai, China, 497–514.

[57] Mendel Rosenblum and John K. Ousterhout. 1992. The Design and Implementation of a Log-Structured File System. *ACM Transactions on Computer Systems* 10, 1 (1992), 26–52.

[58] Justin Sheehy and David Smith. 2010. *Bitcask: A Log-Structured Hash Table for Fast Key/Value Data.* Technical Report. Basho Technologies.

[59] TiKV Project. 2021. AgateDB: A Persistent Key-Value Storage in Rust. Retrieved July 27, 2026 from https://github.com/tikv/agatedb

[60] Stephen Tu, Wenting Zheng, Eddie Kohler, Barbara Liskov, and Samuel Madden. 2013. Speedy Transactions in Multicore In-Memory Databases. In *SOSP*. ACM, Farmington, PA, 18–32.

[61] Alexandre Verbitski, Anurag Gupta, Debanjan Saha, Murali Brahmadesam, Kamal Gupta, Raman Mittal, Sailesh Krishnamurthy, Sandor Maurice, Tengiz Kharatishvili, and Xiaofeng Bao. 2017. Amazon Aurora: Design Considerations for High Throughput Cloud-Native Relational Databases. In *SIGMOD*. ACM, Chicago, IL, 1041–1052.

[62] Gavin Wood. 2014. Ethereum: A Secure Decentralised Generalised Transaction Ledger. Ethereum Project Yellow Paper.

[63] Maofan Yin, Hongbo Zhang, Robbert van Renesse, and Emin Gün Sirer. 2020. CedrusDB: Persistent Key-Value Store with Memory-Mapped Lazy-Trie. arXiv:2005.13762.

[64] Geoffrey X. Yu, Markos Markakis, Andreas Kipf, Per-Åke Larson, Umar Farooq Minhas, and Tim Kraska. 2022. TreeLine: An Update-In-Place Key-Value Store for Modern Storage. *Proceedings of the VLDB Endowment* 16, 1 (2022), 99–112.

[65] Jinghuan Yu, Sam H. Noh, Young-ri Choi, and Chun Jason Xue. 2023. ADOC: Automatically Harmonizing Dataflow Between Components in Log-Structured Key-Value Stores for Improved Performance. In *FAST*. USENIX Association, Santa Clara, CA, 65–80.

[66] Benjamin Zhu, Kai Li, and R. Hugo Patterson. 2008. Avoiding the Disk Bottleneck in the Data Domain Deduplication File System. In *FAST*. USENIX Association, San Jose, CA, 269–282.